

A wave forecasting method based on probabilistic diffusion LSTM network for model predictive control of wave energy converters

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ABSTRACT

Accurate long-term prediction of wave excitation forces is critical for optimizing wave energy converters, enabling enhanced control, and improving energy absorption efficiency. Traditional prediction models often employ deterministic conservative approaches, overlooking uncertainties. This paper introduces a novel prediction method based on the probabilistic diffusion model, offering a more precise prediction while accounting for uncertainty. Notably, our approach goes beyond previous works by simultaneously achieving control objectives for maximizing energy absorption, contrasting with methodologies solely focused on prediction and control tasks. The proposed method utilizes long short-time memory to extract temporal information from historical wave excitation observation. The hidden features are then processed through a diffused probabilistic-based unit. Subsequently, a time-scale neural network is developed for wave excitation moment prediction, followed by the application of nonlinear model predictive control to maximize the energy absorption. The methodology is validated on the 1:20 Wavestar prototype devices through numerical simulations. Results indicate that the predicted excitation moment closely align with measured values. In cases 4, 5, 6, the proposed method yields a substantial improvement in maximum control energy absorption—22%, 16%, and 31%, respectively—compared to traditional prediction methods. The proposed approach not only achieves a more accurate wave excitation moment prediction with uncertainty consideration but also introduces advanced control strategies in a full-scale plant application. This work contributes significantly to the field by bridging the gap between precise prediction and effective control in wave energy conversion systems.

1. Introduction

Wave energy has been recognized as one of the most potential renewable energy resources which are expected to play an incredibly important role in the foreseen energy future [1,2]. A wave energy converter (WEC) is the primary device used to extract energy from the sea. To achieve maximum energy absorption, a lot of effort has gone into developing a control method to control the WEC's motion and the excitation force [3]. When the velocity motion of the WEC accurately matches the phase of the excitation force, the maximum energy extraction is obtained. However, noise can cause the opposite effect, making the WEC an energy exhaustion rather than an energy generator. To prevent this, the control's first priority should be an accurate estimation of the wave excitation force. The next goal should be a strategy that ensures positive economic feedback to make the WEC work as an energy generator [4]. A wave excitation force estimation method based on the float position and angular velocity has been proposed in the literature [5]. The Kalman observer is commonly used for state estimation in WEC devices [6].

In the optimal control of a WEC, the incident wave elevation and wave excitation force are essential to obtain the optimal control input [7]. It is crucial to have a short-term wave excitation force but also beneficial to have long-term information to supervise the control and improve energy absorption [8]. Conventional prediction methods, such as the nonlinear auto-regressive model, have limited short-term prediction performance [9–11]. The time phase lag also limits the maximization of control performance [12,13]. Despite the limitations, many studies still use the nonlinear auto-regressive neural network (NAR) for simplicity in predicting the wave excitation force [4, 14].

The earlier works on wave predictions were mainly based on deterministic prediction. Wave energy conversion (WEC) control requires accurate predictions of wave excitation force and wave elevation. However, conventional prediction methods such as nonlinear auto-aggressive models only provide short-term predictions, ignoring uncertainty information [15]. Some similar shortcomings also existed in [16,17]. To maximize energy absorption, a recently-proposed

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optimal broadband control model [18], which showed better performance than the work in [19,20]. The implementation of the moment-based receding-horizon control method to maximize energy harvest in wave energy converters is reported in [21] and [22], however, the accurate estimation of wave excitation moment is crucial to ensuring optimal controller performance. Some other studies are based on hydrodynamic excitation force estimation and forecasting based on the Kalman filter, which is practically implemented with satisfactory performance [23,24]. Zhan's work also considers the noncasual factor for the wave energy optimal control, however, their methods rely on the accurate state-space model to solve the optimal control input for the PTO device, also with large use of computation resources [25]. An enhanced excitation force estimation approach by incorporating pressure measurement into the feedback control loop is presented in [26]. However, this necessitates the installation of an additional pressure sensor, which increases cost and generates noise simultaneously. [27] inspires us to introduce the nonlinear controller design in the linear system which can improve the performance in the fixed time interval, and also produces a stable solution. Some other exploration about the wave excitation force estimation can be seen in [28–31]. Pena studied the excitation force estimator and made different comparisons for wave-energy devices [32]. Recently, the wave excitation moment prediction is also been the attention of many wave energy converter researchers, to sum up, the real-time requirement of the prediction is becoming more and more important for application improvement, so these intelligent predictors consider the wave excitation moment prediction more effectively [33–36]. Some other relevant discussions about the wave excitation force prediction are given in [37–40].

Predictions of nonlinear wave excitation moments are ubiquitous across various real-world scenarios. [41] investigated the multi-point-absorber case using a semisubmersible floating platform, and it did many numerical studies for the wave converter. They proposed a new arrangement of the Wavestar device and analyzed the time response. However, their work mainly focuses on regular wave simulation, while the practical wave is commonly irregular. The performance of a point absorber wave energy converter was analyzed using a nonlinear numerical model. The model was a numerical time model applied to a three-dimensional point absorber and the results showed the dynamical properties of the hemisphere oscillating system [42]. [43] studied the estimation and forecasting of the wave excitation force to maximize power conversion using the Wavestar device. They utilized an observer-based estimator and a Gaussian process for these purposes. However, their work ignores the long-term non-causality of the required PTO force and wave resource. Yet their work inspires us that the random Gaussian process method is helpful to the wave excitation estimation. Some similar strategies have been shown that the method that considers uncertainty presents comparable or better performance than AR and NNs model approaches [16,44–46]. [47] studied short-term wave forecasting using the Gaussian process for the optimal control of wave energy converters, the predicted values of wave elevation and short-term frequency improved the prediction performances. Recently, as reported in [48], the use of probabilistic Bayesian neural networks for wave elevation prediction has received increased attention due to their ability to address uncertainty.

In the recent decade, with the development of machine learning, the data-driven model has emerged in much literature, providing a good direction for wave energy force prediction [49,50]. In [51], a long short-time memory (LSTM) network for the wave excitation force forecasting, compared to the nonlinear auto-aggressive neural network (NAR) and group method of data handling network. The advantage of their proposed method improves the prediction performance, However, their method does not consider the performance of the WEC control and energy extraction. Gao et al. propose a significant wave height forecasting using hybrid ensemble deep randomized networks with neurons pruning, however, the full-scale framework is ignored [52]. In [53], a dynamic ensemble deep echo state network for significant

wave height forecasting, which considers the flexibility of wave prediction. Recently, the diffusion-based neural network has attracted much attention [54,55]. The diffusion process introduces the Gaussian noise to the original distribution and then achieves the variants of the feature information, which has been widely in the vision and text tasks [56,57]. The use of normalizing flows for the continuous system by adding uniform noise to the data can penalize the probability distribution since they can consider the uncertainty in the learning process. Several recent contributions in the literature are presented in Table 1.

This work focuses on the development of a novel wave excitation moment prediction method that can address these limitations. Different from the previous deterministic works, this work explores the implementation of the probabilistic sequence model, which can quantify the uncertainty during the wave prediction process. In particular, this work investigates, for the first time, the diffusion probabilistic sequence neural network, for the wave excitation moment prediction. The improved probabilistic diffused neural network implements a more accurate prediction and considers the uncertainty. Additionally, our work goes beyond the previous works which only focused on wave prediction by implementing a nonlinear real-time control strategy, to the author's best knowledge, it is also the first time. Therefore, the main bullets of this paper are summarized as:

- Comparing to traditional conservative wave prediction methods [42–45], a novel prediction model based on a probabilistic diffused neural network is proposed. The proposed prediction leverages the information of uncertainty to learn a posterior distribution of the wave, thus predicting the wave excitation moment. In the presented prediction model, the sequence information is processed by a diffused process to extract the different distributions.
- Diverging from conventional deterministic prediction methods employed in the past [25,27,30–33], not limited by the prediction consideration, the presented a model that implements a wave excitation moment prediction, the prediction provides valuable information reference for the later nonlinear model predictive control. Therefore, a full-scale process that includes prediction and control is presented in this paper.
- The proposed method is developed based on the probabilistic diffusion approach, which is very novel in the research cutting-edge and can fully take advantage of both the sequence machine learning model's ability in tackling complex nonlinear problems and probabilistic diffusion's uncertainty quantification merits. Based on the proposed accurate prediction, we further implement the control strategy with better performance in maximizing energy extraction.
- Developing from the previous diffusion probabilistic models [52–58], control works [59,60], the presented work implements a more accurate prediction and control with maximizing energy absorption in the wave energy converter. The practical comparison of control performance with nonlinear model predictive control is evaluated in the applied Wavestar device, the simulation results show that the energy absorption improved by 22%, 16%, and 31% with 3 irregular wave benchmarks.

The following of this paper is organized as follows. Section 2 introduces some related problem discussions that are the foundation of our research. Section 3 gives the methodology that formulates our diffused long-short time memory prediction method. Section 4 presents the wave excitation moment prediction results and its control performance. Finally, Section 5 concludes the paper and gives some future exploration.

2. Problem description

2.1. Overall control system

The whole framework of the wave energy converter plant is given in Fig. 1. The wave elevations with time sequence are sourced from the sea, which is transformed to the wave excitation force which is

Table 1
State-of-the-art literature review for the wave prediction and control.

Ref	Prediction/Control	Robustness	Full-scale	Uncertainty	Generalization
[58]	predictive-only	Yes	No	deterministic	Feasible
[53]	predictive-only	Yes	Yes	deterministic	Infeasible
[52]	predictive-only	Yes	No	deterministic	Infeasible
[10]	control-only	No	No	deterministic	Infeasible
[48]	control-only	Yes	No	probabilistic	Feasible
[51]	predictive-only	No	No	deterministic	Infeasible
[46]	predictive-only	No	No	deterministic	Infeasible
[45]	control-only	No	No	deterministic	Infeasible
[59]	control-only	Yes	No	probabilistic	Infeasible
[44]	predictive-only	No	No	deterministic	Infeasible
[43]	predictive-only	No	No	deterministic	Infeasible
[60]	control-only	No	No	probabilistic	Infeasible
[40]	predictive-only	No	No	deterministic	Infeasible
[39]	predictive-only	No	No	deterministic	Infeasible
[38]	predictive-only	Yes	No	deterministic	Infeasible
[54]	predictive-only	Yes	No	probabilistic	Infeasible
[24]	predictive-only	Yes	Yes	deterministic	Feasible
[37]	control-only	No	No	deterministic	Infeasible
[36]	predictive-only	Yes	Yes	deterministic	Infeasible
[32]	predictive-only	No	No	deterministic	Infeasible
[28]	predictive-only	No	No	probabilistic	Infeasible
[25]	control-only	No	No	deterministic	Infeasible
[21]	control-only	Yes	No	deterministic	Infeasible
[16]	predictive-only	Yes	Yes	deterministic	Feasible
[15]	control-only	Yes	No	deterministic	Infeasible
Ours	prediction & control	Yes	Yes	probabilistic	Feasible

denoted as $F_e(t)$. The excitation force purges the hydrodynamic float motion with the phase of sea waves, and the power-take-off (PTO) device extracts energy at the same time. The measured variables by the multiple sensors from the WEC plant model are the float position $x(t)$, velocity $\dot{x}(t)$, acceleration $\ddot{x}(t)$, and the force output from the PTO $F_{pto}(t)$. Due to this reason, the wave excitation moment cannot be directly measured, so a common way to solve this is to define the wave excitation moment as a state variable and then use the Kalman estimator to approximate it. The wave excitation moment under the current instant t is estimated by the measurable float position, velocity, and acceleration defined by $F_{exc}(t)$. The next stage is to consider the strategy to make the PTO obtain the maximum energy from the sea, the core principle behind this is to make the PTO move with the same frequency as the wave elevation.

The main objective of a PTO system is to extract as much energy as possible from the sea waves to the grid for a broad range of sea states. The electrical energy E_e absorbed by the grid over a time horizon T is defined as:

$$E_e = - \int_t^{t+T} P_e(\tau) d\tau = - \int_t^{t+T} \Gamma(\tau) P_m(\tau) d\tau \quad (1)$$

where P_e denotes the electrical power delivered to the grid, P_m is the instantaneous hydromechanical power absorbed by the PTO system, the overall efficiency of the PTO system, and τ is the variable of integration. The negative sign in (1) is because the energy is drawn from the WEC and thus, the maximization of the energy absorbed corresponds to a minimization of the control objective [11]. The instantaneous hydromechanical absorbed power is given by:

$$P_m(t) = F_{pto}(t) \dot{\theta}(t) \quad (2)$$

where F_{pto} is the PTO moment and θ represents the angular velocity of the arm. To use standard nomenclature, $F_{pto}(t_k)$ is replaced by u_k . Substituting (2) in (1) the optimal control problem can be formulated as:

$$\min_u - E_e = \int_t^{t+I} \Gamma(\tau) u(\tau) \dot{\theta}(\tau) d\tau \quad (3)$$

$$\begin{aligned} \min_u - E_e &= \int_t^{t+I} \Gamma(\tau) u(\tau) \dot{\theta}(\tau) d\tau \\ s.t. \quad &\begin{cases} x_{k+1} = f(x_k, u_k, w_k) \\ U_{\min} \leq u_k \leq U_{\max} \end{cases} \end{aligned} \quad (4)$$

The equivalent discrete-time optimization problem is given by:

$$\begin{aligned} \min_{ui} J &= \sum_{i=1}^{N_p} \gamma_{k+i} u_{k+i-1} \dot{\theta}_{k+i} \\ s.t. \quad &\begin{cases} x_{k+1} = f(x_k, u_k, w_k) \\ U_{\min} \leq u_k \leq U_{\max} \end{cases} \end{aligned} \quad (5)$$

where N_p is the prediction horizon. Eq. (4)b represents the WEC dynamics, with the state $x = [\theta, \dot{\theta}, r]^T$, u_k is the control input, w_k the discrete-time value for the excitation moment M_{exc} , and γ_k the specific value for the PTO efficiency at time instant t_k , that is, $\Gamma(t_k) = \gamma_k$. Due to the reason that the output from the Kalman estimator is only the current instant excitation moment, the constrained controller needs to consider the further time step, so a wave excitation moment prediction model is critical before it is transformed to the control unit to maximize the energy maximum objective.

2.2. Wave excitation moment forecasting

In the WEC-Sim general model, the following architecture is used. The usual strategy of the framework can be divided into three categories, estimation, prediction, and control. In the estimation module, due to many of the optimal variables of the wave energy converter being hard to measure by the physical device, however, the reliability of the system is largely related to the availability of measurements of the wave excitation force or excitation moment caused by the incoming waves. In practical scenarios, it is hard to deploy such a lot of sensors in the underwater environment, which will cause high costs and inaccuracy. Therefore, the wave excitation force should be estimated via some other process variables, such as the position and velocity of the float.

Many alternatives to this problem have been proposed by the literature (ref to Table 1). A commonly used method is the Kalman filter coupled with a random-walk model for the estimation of wave excitation force. Some control-domain researcher also extends this to the bode-stability analysis and then use the system transfer function to compensate for the state, which produces a satisfactory approximation of the wave excitation moment.

Another core element in the Wavestar module is the wave excitation prediction, the source from the estimation should be given as the input of some prediction algorithm for obtaining the control input. Much

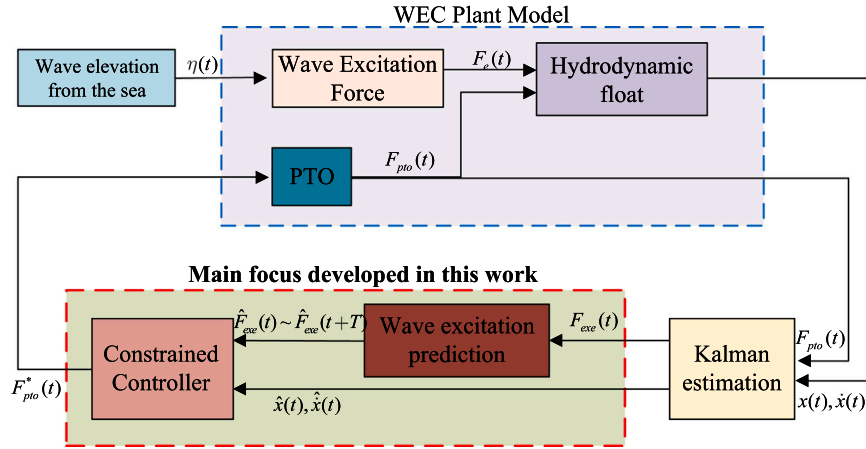


Fig. 1. The full framework of the wave energy conversion system.

literature is based on the auto-aggressive model (AR) to predict the wave excitation moment. As given in [15,16,18], AR-based methods imply that the wave excitation moment $M_{exc,k}$ at any given time instant t_k is linearly independent of its past values via the AR coefficient a_i . However, the practical case demonstrates that the information from the waves is a time-sequence series, which means that the successor information is linked to the historical front point. The AR-based method avoids this view and would cause a lot of information loss during the approximation, which would affect the prediction accuracy of the wave excitation moment. On the other hand, the AR prediction has a large time delay which directly degrades the prediction accuracy and decreases the control performance simultaneously. In many control strategies of PTO devices, the uncertainty description of prediction information of the variable should be helpful to improve prediction accuracy, thus improving the energy absorption efficiency, which motivates us to build an improved model to focus on wave excitation prediction. Therefore, under the constraints of the general energy maximization objection, the wave excitation moment prediction can be written as:

Given: The historical time sequence wave excitation moment series denoted as

$$\mathbf{x}_{t-1} = \langle F_{exc}(t-m-1), F_{exc}(t-m), \dots, F_{exc}(t-1) \rangle$$

where t is the historical time window instant. m is the historical total time interval for the further wave excitation moment prediction.

Predict: The future time sequence wave excitation moment denoted as

$$\mathbf{x}_t = \langle F_{exc}(t), F_{exc}(t+1), \dots, F_{exc}(t+N_p) \rangle$$

where $N_p = T - t + 1$ is the preceding horizon that is used as the nonlinear controller to obtain the maximum energy.

The simplest way to implement the above goal is the autoaggressive model, which also has been used in many WECs. However, the uncertainty in their model has not been quantified.

2.3. Wavestar modeling

In this testing case, the selected device is the Wavestar scaled model employed in the WEC control competition. Wavestar is a model used in the WEC Control Competition (WECCOMP) which has been studied in many kinds of literature. For research brevity, the model is set as the minimum of the real device. As shown in Fig. 1, a semi-sphere is served as the main floater which extracts the energy simultaneously from the experiment wave gauges. The floater is connected to a rotating arm, which is hinged at a fixed reference point (Point A). Based on an assumption that all the system oscillations are very small that can be

ignored, then the dynamic of the equation of motion can be written as:

$$(J + J_\infty) \ddot{\theta}(t) = -K_{hs}\theta(t) - b_v\dot{\theta}(t) - M_r(t) - M_{exc}(t) + M_{pto}(t) \quad (6)$$

$$\dot{\theta}(t) = A_r(t) + B_r\dot{\theta}(t) \quad (7)$$

$$M_r(t) = C_r(t) + D_r\dot{\theta}(t) \quad (8)$$

where:

1. θ is the angular displacement of the arm with respect to the equilibrium position, $\dot{\theta}$ and $\ddot{\theta}$ represent the angular velocity and angular acceleration of the arm.
2. J is the total mass moment of the inertia of the float and pivot arm.
3. J_∞ is the added mass moment of the inertia.
4. K_{hs} is the hydrostatic coefficient,
5. b_v is a linear damping coefficient
6. M_r is the radiation moment.
7. M_{exc} is the excitation moment due to the incident wave
8. M_{pto} is the PTO moment (input to the system).
9. (A_r, B_r, C_r, D_r) are the state space matrices used to approximate the radiation moment M_r avoiding the direct computation of the convolution integral in time-domain simulation. The state vector has no physical meaning but still contains information on the condition of the surrounding fluid [16].

3. Methodology

3.1. Wave excitation moment forecasting

Based on the problem formulation, our target is to address the wave excitation moment forecasting based on the historical discrete time instant $m-1, m, t+1, \dots, t-1$, and predict the wave excitation moment at time $t, t+1, \dots, t+T$ with N_p receding horizon. As given in Fig. 2, the measurements from the multiple sensors obtain the historical wave variables which contain the position, velocity, and acceleration, these variables can be denoted as:

$$\mathbf{x}_h(t) = \begin{bmatrix} F_{exc}(t-h) \\ F_{exc}(t-h-1) \\ \vdots \\ F_{exc}(t-h-m) \\ F_{exc}(t-h-m-1) \end{bmatrix} \quad (9)$$

and its pto force denoted as $F_{pto}(t)$. The past observation of wave excitation moment with $m+1$ steps proceeds to the ML model with

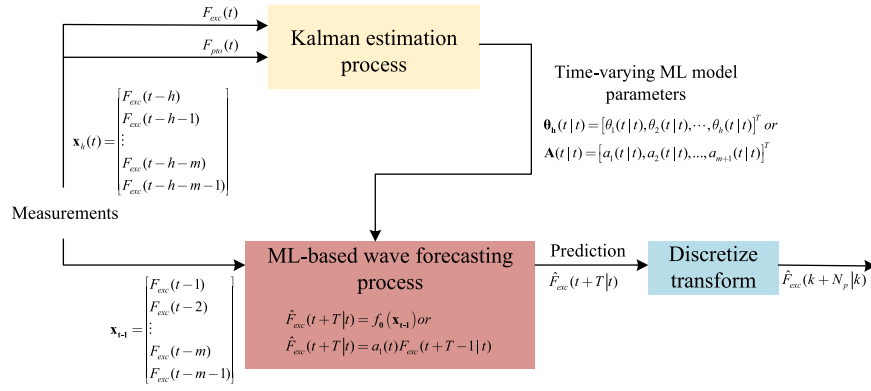


Fig. 2. Synoptic illustration of ML-based wave excitation moment forecasting problem.

time-varying hyper-parameters $\theta_h(t|t)$ or $A(t|t)$. The problem can then be formulated to optimize the ML-based model and find the optimal coefficients. The N_p steps multi-ahead wave excitation moment prediction is denoted in a discrete form $\hat{F}_{exc}(t+T|t)$. The ML-based prediction model can be given by:

$$\begin{aligned} \hat{F}_{exc}(t+T|t) &= f_{\theta}(\mathbf{x}_{t-1}) \text{ or} \\ \hat{F}_{exc}(t+T|t) &= \mathbf{A}(t)\mathbf{x}_{t-1}^T \end{aligned} \quad (10)$$

where θ is the hyper-parameters, $\mathbf{A}$ is the coefficient with historical approximation and f is the nonlinear mapping function. For this purpose, the data-driven machine learning prediction methods are first reviewed. The commonly used autoaggressive method is given and will be compared to the new approaches developed in this paper. These methods are commonly machine learning-based (ML-based) approaches.

3.2. Auto-regressive model-based approach

The objective of wave excitation moment developed in this paper is to predict the future $\mathbf{x}_t$ based on the historical time series $\mathbf{x}_{t-1}$. The AR approach is built on the assumption that $\mathbf{x}_t$ at time instant t is a linear approximation of its past observation $F_{exc}(t-1), F_{exc}(t-2), \dots, F_{exc}(t-m), F_{exc}(t-m-1)$:

$$F_{exc}(t) = a_1 F_{exc}(t-1) + a_2 F_{exc}(t-2) + \dots + a_m F_{exc}(t-m) + a_{m+1} F_{exc}(t-m-1) + v(t) \quad (11)$$

where $(a_1, a_2, \dots, a_{m+1})$ are the parameters. $m+1$ is the order of the auto-regressive model, and $v(t)$ is the zero-mean white noise distribution. Consider that coefficients $a_i, \forall i = 1, 2, 3, \dots, m, m+1$ at time t are already calculated. So the best prediction of $\hat{F}_{exc}(t+T|t)$ at time t is:

$$\begin{aligned} \hat{F}_{exc}(t+T|t) &= a_1(t)F_{exc}(t+T-1|t) \\ &+ a_2(t)F_{exc}(t+T-2|t) + \dots + a_m(t)F_{exc}(t+T-m|t) \\ &+ a_{m+1}(t)F_{exc}(t+T-m-1|t) \end{aligned} \quad (12)$$

where $\hat{F}_{exc}(t+T-i|t) = F_{exc}(t+T-i)$ if $t+T-i \leq t$ or $t-i \leq 0$.

Eq. (11) can be rewritten as a compact vector form as:

$$\mathbf{x}_t = \mathbf{x}_{t-1}^T \mathbf{A} + \mathbf{v}(t) \quad (13)$$

where $\mathbf{A} = [a_1, a_2, \dots, a_{m+1}]^T$ and

$$\mathbf{x}_{t-1} = [F_{exc}(t-1), F_{exc}(t-2), \dots, F_{exc}(t-m), F_{exc}(t-m-1)]^T.$$

The solution target of (11) is to obtain the coefficients vector $\mathbf{A}_i, i = 1, 2, \dots, m+1$ at time t , it can be obtained by minimizing the following multi-step long-range accumulation ahead error:

$$\min_{a_1, a_2, \dots, a_{m+1}} \sum_{i=m+N_p+1}^t \sum_{j=1}^{N_p} (F_{exc}(t) - \hat{F}_{exc}(t|t-i))^2 \quad (14)$$

where N_p is the forecasting horizon over the AR model. After the optimization training process, the AR-based model can predict the

preceding N_p steps wave excitation moments based on the historical $m+1$ past observation steps.

3.3. Long-short time memory

Recurrent neural networks (RNNs) have been recognized to achieve superior performance in solving time-instant sequence problems. A widely accepted shortcoming of RNNs is the long-term dependency and vanishing gradient problem. A special kind of RNNs called long short-time memory (LSTM) is capable of learning long-term dependencies. In the basic LSTM, three gates to protect and control the cell state and some repeating modules are added in an LSTM containing four interacting layers. Denote the input vector $x(t)$ and the previous hidden state $h(t-1)$, and the external inputs are its previous cell state $c(t-1)$. Then, the forget gate is triggered as:

$$f(t) = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (15)$$

And the input gate and new candidate vectors are computed as follows:

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (16)$$

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \quad (17)$$

The new cell state in LSTM is updated by the following equation.

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t \quad (18)$$

The output gate vector can be calculated as follows:

$$o_t = \sigma(W_o [h_{t-1}, x_t] + b_o) \quad (19)$$

$$h_t = o_t * \tanh(C_t) \quad (20)$$

In Eqs. (3)–(8), where σ is the nonlinear activation function, usually, the Sigmoid function. Tanh represents the nonlinear tangent activation function. $*$ represents the point-wise multiplication operation. W_c, W_o, W_i are the related weights and b_i, b_c, b_o are the bias, respectively. In the LSTM unit, the tangent activation can be replaced with other activation kernels because Tanh is used for state and output, for processing data, and this can also be done with other activation functions. The Sigmoid function is used between the different gates to project the output into the range of 0 to 1.

3.4. Diffusion model forecasting

Let $\mathbf{x}^0 \sim q_X(\mathbf{x}^0)$ denote the multivariate training vector from some input space $\mathcal{X} = \mathbb{R}^D$ and let $p_{\theta}(\mathbf{x}^0)$ denote the probability density function (PDF). Diffusion models are latent variable models of the form

$p_\theta(\mathbf{x}^0) := \int p_\theta(\mathbf{x}^{0:N}) d\mathbf{x}^{1:N}$, where $\mathbf{x}^1, \dots, \mathbf{x}^N$ are latent of dimension $\mathbb{R}^D$. The variational approximate posterior $q(\mathbf{x}^{1:N} | \mathbf{x}^0)$ is:

$$q(\mathbf{x}^{1:N} | \mathbf{x}^0) = \prod_{n=1}^N q(\mathbf{x}^n | \mathbf{x}^{n-1}) \quad (21)$$

The above equation is not trainable but fixed to a Markov chain (called the forward process) that gradually adds Gaussian noise to the hidden representation:

$$q(\mathbf{x}^n | \mathbf{x}^{n-1}) := \mathcal{N}(\mathbf{x}^n; \sqrt{1 - \beta_n} \mathbf{x}^{n-1}, \beta_n \mathbf{I}). \quad (22)$$

The forward process uses an increasing variance schedule $\beta_1, \dots, \beta_N$ with $\beta_n \in (0, 1)$. The joint distribution $p_\theta(\mathbf{x}^{0:N})$ is called the reverse process and is defined as a Markov chain with learned Gaussian transitions starting with $p(\mathbf{x}^N) = \mathcal{N}(\mathbf{x}^N; \mathbf{0}, \mathbf{I})$, where each subsequent transition of

$$p_\theta(\mathbf{x}^{0:N}) := p(\mathbf{x}^N) \prod_{n=N}^1 p_\theta(\mathbf{x}^{n-1} | \mathbf{x}^n) \quad (23)$$

is given by a parametrization of our choosing denoted by

$$p_\theta(\mathbf{x}^{n-1} | \mathbf{x}^n) := \mathcal{N}(\mathbf{x}^{n-1}; \mu_\theta(\mathbf{x}^n, n), \Sigma_\theta(\mathbf{x}^n, n) \mathbf{I}) \quad (24)$$

with shared parameters θ . Both $\mu_\theta : \mathbb{R}^D \times \mathbb{N} \rightarrow \mathbb{R}^D$ and $\Sigma_\theta : \mathbb{R}^D \times \mathbb{N} \rightarrow \mathbb{R}^+ \text{ take two inputs, namely the variable } \mathbf{x}^n \in \mathbb{R}^D \text{ as well as the noise index } n \in \mathbb{N}. \text{ The goal of } p_\theta(\mathbf{x}^{n-1} | \mathbf{x}^n) \text{ is to eliminate the Gaussian noise added in the diffusion process. The parameters } \theta \text{ are learned to fit the data distribution } q_{\mathcal{X}}(\mathbf{x}^0) \text{ by minimizing the negative loglikelihood via a variational bound using Jensen's inequality:}$

$$\min_\theta \mathbb{E}_{q(\mathbf{x}^{0:N})} [-\log p_\theta(\mathbf{x}^0)] \leq \min_\theta \mathbb{E}_{q(\mathbf{x}^{0:N})} [-\log p_\theta(\mathbf{x}^{0:N}) + \log q(\mathbf{x}^{1:N} | \mathbf{x}^0)]. \quad (25)$$

This upper bound can be shown to be equal to

$$\min_\theta \mathbb{E}_{q(\mathbf{x}^{0:N})} \left[-\log p(\mathbf{x}^N) - \sum_{n=1}^N \log \frac{p_\theta(\mathbf{x}^{n-1} | \mathbf{x}^n)}{q(\mathbf{x}^n | \mathbf{x}^{n-1})} \right]. \quad (26)$$

As given in [61], a property of the forward process is that it admits sampling $\mathbf{x}^n$ at any arbitrary noise level n in closed form, since if $\alpha_n := 1 - \beta_n$ and $\bar{\alpha}_n := \prod_{i=1}^n \alpha_i$ its cumulative product, we have:

$$q(\mathbf{x}^n | \mathbf{x}^0) = \mathcal{N}(\mathbf{x}^n; \sqrt{\bar{\alpha}_n} \mathbf{x}^0, (1 - \bar{\alpha}_n) \mathbf{I}). \quad (27)$$

By using the fact that these processes are Markov chains, the objective in (2) can be written as the KL-divergence between Gaussian distributions:

$$-\log p_\theta(\mathbf{x}^0 | \mathbf{x}^1) + D_{\text{KL}}(q(\mathbf{x}^N | \mathbf{x}^0) \| p(\mathbf{x}^N)) + \sum_{n=2}^N D_{\text{KL}}(q(\mathbf{x}^{n-1} | \mathbf{x}^n, \mathbf{x}^0) \| p_\theta(\mathbf{x}^{n-1} | \mathbf{x}^n)) \quad (28)$$

and shows that by the property (3) the forward process posterior in these KL divergences when conditioned on $\mathbf{x}^0$, i.e. $q(\mathbf{x}^{n-1} | \mathbf{x}^n, \mathbf{x}^0)$ are tractable given by

$$q(\mathbf{x}^{n-1} | \mathbf{x}^n, \mathbf{x}^0) = \mathcal{N}(\mathbf{x}^{n-1}; \tilde{\mu}_n(\mathbf{x}^n, \mathbf{x}^0), \tilde{\beta}_n \mathbf{I}) \quad (29)$$

where

$$\tilde{\mu}_n(\mathbf{x}^n, \mathbf{x}^0) := \frac{\sqrt{\bar{\alpha}_{n-1}} \beta_n}{1 - \bar{\alpha}_n} \mathbf{x}^0 + \frac{\sqrt{\alpha_n} (1 - \bar{\alpha}_{n-1})}{1 - \bar{\alpha}_n} \mathbf{x}^n \quad (30)$$

and

$$\tilde{\beta}_n := \frac{1 - \bar{\alpha}_{n-1}}{1 - \bar{\alpha}_n} \beta_n. \quad (31)$$

Further, [61] shows that the KL-divergence between Gaussians can be written as:

$$D_{\text{KL}}(q(\mathbf{x}^{n-1} | \mathbf{x}^n, \mathbf{x}^0) \| p_\theta(\mathbf{x}^{n-1} | \mathbf{x}^n)) = E_q \left[\frac{1}{2\Sigma_\theta} \left\| \tilde{\mu}_n(\mathbf{x}^n, \mathbf{x}^0) - \mu_\theta(\mathbf{x}^n, n) \right\|^2 \right] + C \quad (32)$$

where C is a constant which does not depend on θ . So instead of a parametrization (1) of p_θ that predicts $\tilde{\mu}$, one can instead use the property (3) to write $\mathbf{x}^n(\mathbf{x}^0, \epsilon) = \sqrt{\bar{\alpha}_n} \mathbf{x}^0 + \sqrt{1 - \bar{\alpha}_n} \epsilon$ for $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and

the formula for $\tilde{\mu}$ to obtain that μ_θ must predict $(\mathbf{x}^n - \beta_n \epsilon / \sqrt{1 - \bar{\alpha}_n}) / \sqrt{\alpha_n}$, but since $\mathbf{x}^n$ is available to the network, we can choose:

$$\mu_\theta(\mathbf{x}^n, n) = \frac{1}{\sqrt{\alpha_n}} (\mathbf{x}^n - \frac{\beta_n}{\sqrt{1 - \bar{\alpha}_n}} \epsilon_\theta(\mathbf{x}^n, n)) \quad (33)$$

where ϵ_θ is a network which predicts $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ from $\mathbf{x}^n$, then the objective simplifies to:

$$\mathbb{E}_{\mathbf{x}^0, \epsilon} \left[\frac{\beta_n^2}{2\Sigma_\theta \alpha_n (1 - \bar{\alpha}_n)} \|\epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_n} \mathbf{x}^0 + \sqrt{1 - \bar{\alpha}_n} \epsilon, n)\|^2 \right] \quad (34)$$

which resembles the loss in noise conditional score networks using score matching [62]. The reverse process $\mathbf{x}^{n-1} \sim p_\theta(\mathbf{x}^{n-1} | \mathbf{x}^n)(1)$ is computed by:

$$\mathbf{x}^{n-1} = \frac{1}{\sqrt{\alpha_n}} (\mathbf{x}^n - \frac{\beta_n}{\sqrt{1 - \bar{\alpha}_n}} \epsilon_\theta(\mathbf{x}^n, n)) + \sqrt{\Sigma_\theta} \mathbf{z} \quad (35)$$

where $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ for $n = N, \dots, 2$ and $\mathbf{z} = \mathbf{0}$ when $n = 1$. The full sampling procedure for $\mathbf{x}^0$. The ultimate output involves recursively sampling N times from the learned resembles Langevin dynamic distribution.

3.5. Diffused-LSTM training process

We denote the entities of a multivariate time series by $\mathbf{x}_{i,t}^0 \in \mathbb{R}$ for $i \in \{1, \dots, D\}$ where t is the time index. Thus the multivariate vector at time t is given by $\mathbf{x}_t^0 \in \mathbb{R}^D$. We are tasked with predicting the multivariate distribution of some given prediction time steps into the future and so in what follows consider time series with $t \in [1, T]$, sampled from the complete time series history of the training data, where we will split this contiguous sequence into a context window of size $[1, t_0]$ and prediction interval $[t_0, T]$, reminiscent of seq-to-seq models in language modeling [63].

Over the prediction window of an individual time series, the univariate probabilistic model maximizes the log-likelihood of each entity $\mathbf{x}_{i,t}^0$ at a time step $t \in [t_0, T]$. This is done in relation to the parameters of a selected distributional model by using the state of an LSTM derived from its previous time step $\mathbf{x}_{i,t-1}^0$ and its corresponding covariates $\mathbf{c}_{i,t-1}$. The emission distribution model is chosen to best match the statistics of the time series, which is typically Gaussian for real-valued data and negative binomial for count data, and the network incorporates activation functions that satisfy the constraints of the distribution's parameters; for the scale parameter of the Gaussian, a *softplus* activation is used [64].

As shown in [54], TimeGrad is used to learn a model of the conditional distribution of the future time steps of a univariate time series given its past and covariates as:

$$q_{\mathcal{X}}(\mathbf{x}_{t_0:T}^0 | \mathbf{x}_{1:t_0-1}^0, \mathbf{c}_{1:T}) = \prod_{t=t_0}^T q_{\mathcal{X}}(\mathbf{x}_t^0 | \mathbf{x}_{1:t-1}^0, \mathbf{c}_{1:T}) \quad (36)$$

Assuming known covariates for all time points, each factor is learned through a conditional denoising diffusion model introduced earlier. Modeling the temporal dynamics, we utilize the autoregressive recurrent neural network (RNN) architecture from [63]. This architecture employs LSTM or Gated Recurrent Unit (GRU) to encode the time series sequence up to time point t , considering the covariates $\mathbf{c}_t$ and updating the hidden state as $\mathbf{h}_t$:

$$\mathbf{h}_t = \text{LSTM}_\theta(\text{concatenate}(\mathbf{x}_t^0, \mathbf{c}_t), \mathbf{h}_{t-1}) \quad (37)$$

where LSTM_θ is a multi-layer LSTM parameterized by shared weights θ and $\mathbf{h}_0 = \mathbf{0}$. Thus we can approximate (8) by the model:

$$\prod_{t=t_0}^T p_\theta(\mathbf{x}_t^0 | \mathbf{h}_{t-1}), \quad (38)$$

where now θ comprises the weights of the LSTM as well as the denoising diffusion model. This model is a probabilistic regressive model as it consumes the observations at the time step $t-1$ as input to learn the distribution of, or sample, the next time step as shown in Fig. 3.

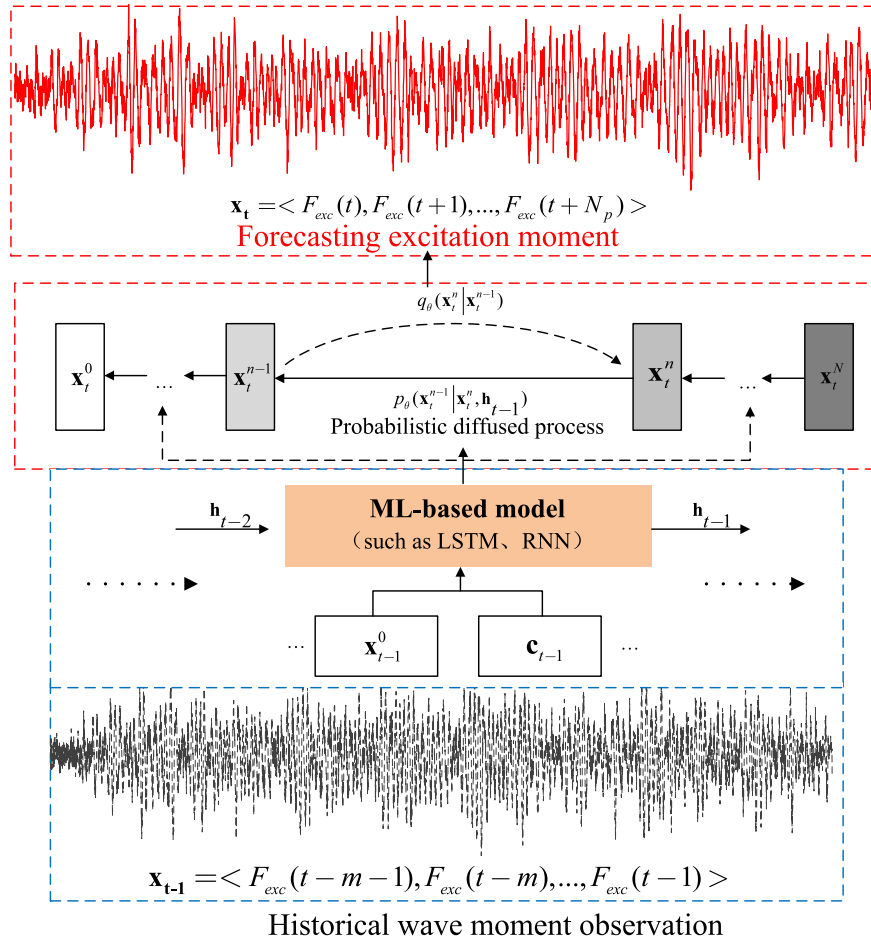


Fig. 3. The framework of the proposed diffusion ML-based wave excitation moment prediction. An ML-based temporal conditioned diffusion probabilistic model at some time $t-1$ depicting the fixed forward that adds Gaussian noise and the learned reverse processes.

Algorithm 1: Diffused probabilistic network training process for each time steps $t \in [t_0, T]$.

Input: historical wave observation $x_t^0 \sim q_X(x_t^0)$ and state h_{t-1} ;

Output: LSTM network parameters θ , z distribution parameters β .

repeat

 Initialize $n \sim \text{Uniform}(1, \dots, N)$ and $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$

 Take gradient on step $\nabla_\theta \|\epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_n}x_t^0 + \sqrt{1 - \bar{\alpha}_n}\epsilon, h_{t-1}, n)\|^2$

 Update the gradient with different optimizer

until;

3.6. Inference and forecasting

Training is performed by randomly sampling wave excitation moment sequences and adjoining prediction-sized windows from the training time series data and optimizing the parameters θ that minimize the negative log-likelihood of the model (10):

$$\sum_{t=t_0}^T -\log p_\theta(x_t^0 | \mathbf{h}_{t-1}) \quad (39)$$

where the hidden state $\mathbf{h}_{t-1}$ is obtained by running the LSTM on the chosen time-series wave excitation moment window. Through a parallel derivation to the one conducted in the preceding section, the conditional form of the objective (4) for time step t and noise index n is expressed as detailed in [61]:

$$\mathbb{E}_{x_t^0, \epsilon, n} [\|\epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_n}x_t^0 + \sqrt{1 - \bar{\alpha}_n}\epsilon, h_{t-1}, n)\|^2] \quad (40)$$

When opting for the variance in (1) to be $\Sigma_\theta = \tilde{\beta}_n(5)$, the ϵ_θ network is now conditioned on the hidden state. Algorithm 1 is the training procedure for each time step in the prediction window using this objective.

Algorithm 2: Sampling wave excitation moment forecasting $x_t^{N_p}$ via annealed Langevin dynamics.

Input: uniform noise $x_{t-T+1}^{N_p} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, distribution parameter β , and state $\mathbf{h}_{t-1}$;

for $n = N_p$ **to** 1 **do**

if $n > 1$ **then**

$z \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$

else

$z = 0$

end

$$x_t^{n-1} = \frac{1}{\sqrt{\alpha_n}}(x_t^0 - \frac{\beta_n}{\sqrt{1-\alpha_n}}\epsilon_\theta(x_t^n, \mathbf{h}_{t-1}, n)) + \sqrt{\Sigma_\theta}z$$

end

Return: $x_t^{N_p}$

After training, the aim is to predict the wave excitation moment for each time series in our data set some prediction steps into the future and compare them with the corresponding test set time series. During the training phase, the LSTM is executed over the last $m+1$ window of the training set to derive the hidden state $\mathbf{h}_T$ using Eq. (9). Subsequently, the sampling procedure outlined in Algorithm 2 is employed to generate a sample x_{T+1}^0 for the next time step. This sample is autoregressively passed to the LSTM along with the covariates $\mathbf{c}_{T+1}$ to obtain the

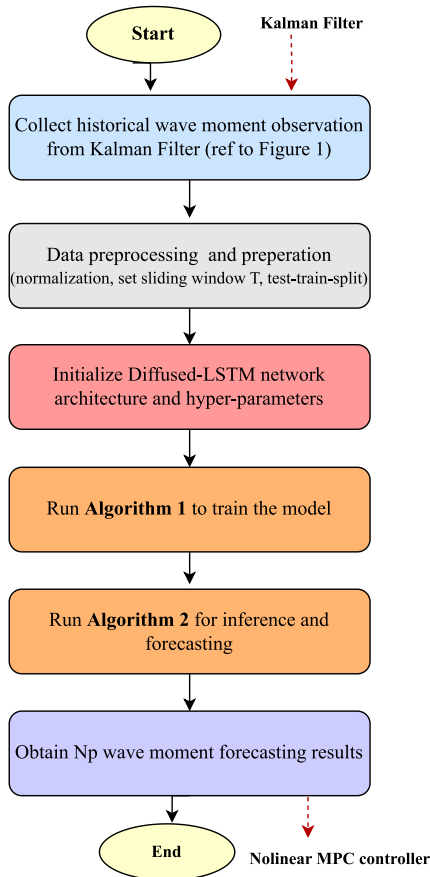


Fig. 4. The whole flowchart of the proposed diffusion ML-based wave excitation moment prediction.

subsequent hidden state $\mathbf{h}_{T+1}$. This process is iteratively repeated until the desired forecast horizon N_p is achieved. This process of sampling trajectories from the “warm-up” state $\mathbf{h}_T$ can be repeated many times (e.g. $S = 100$) to obtain empirical quantiles of the uncertainty of our predictions. Hence, Fig. 4 depicts the flowchart of the Diffused-LSTM model for wave excitation moment prediction.

3.7. Nonlinear model predictive control

The common wave energy converter is a nonlinear system for many practical scenarios. NMPC is becoming increasingly popular for solving such problems. Due to its ability to deal with constraints and nonlinear dynamics, the NMPC has been applied to many domains. Developed from [65], a general nonlinear wave energy converter in a continuous system is given as:

$$\begin{aligned} \dot{x}(t) &= f(x(t), u(t), F_{exc}(t), t) \\ y(t) &= g(x(t), u(t), F_{exc}(t), t), t \in [0, T] \end{aligned} \quad (41)$$

The discretion of (41) is written as:

$$\begin{aligned} x_{k+1} &= f(x_k, u_k, F_{exc}(k)) \\ y_k &= g(x_k, u_k, F_{exc}(k)) \end{aligned} \quad (42)$$

Where x_{k+1} , u_k , and y_k are vectors containing the n_x states, n_u inputs and n_y the outputs of the system, respectively. The output function $g(x_k, u_k, F_{exc}(k))$ is selected as the follows:

$$y_k = g_k = \begin{bmatrix} \dot{\theta}_k \\ \gamma_k u_{k-1} \end{bmatrix} \quad (43)$$

Remark 1. In the practical implementation, the deviation linearization of the above equation needs to consider the wave excitation moment

at time $t = k$ respectively, the excitation moment used in this stage is predicted by the probabilistic diffused model.

As shown in [66], the cost function of the wave energy converter model can be written as:

$$J = \frac{1}{2} \hat{Y} Q \hat{Y} + \frac{1}{2} \delta \hat{U}^T R \delta \hat{U} \quad (44)$$

where the matrix R is a positive definite matrix with dimensions $N_p n_u \times N_p n_u$ and constant elements over its diagonal. Q is selected as a block diagonal matrix with dimensions $N_p n_u \times N_p n_u$ and inner matrices q_i used to compute the product $\dot{\theta}_k \gamma_k u_{k-1}$ is defined in [66].

$$Q = \begin{pmatrix} q_1 & 0 & \cdots & 0 \\ 0 & q_2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & q_{N_p} \end{pmatrix}, q_i = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \forall i = [1, N_p] \quad (45)$$

Equation (18) includes an additional term that penalizes the input deviation. This term is included for two reasons: first, it smooths out the control signal, making the requirement for the actuator's response limit less stringent; and second, according to [10,22,32,43], a reactive control strategy with a cost function solely based on maximizing the extracted energy can result in overall negative energy absorbed, implying that the system is losing energy rather than absorbing energy from the waves. the final standard quadratic programming (QP) formulation is obtained:

$$J = \frac{1}{2} \delta \hat{U}^T E \delta \hat{U} + \delta \hat{U}^T f \quad s.t. \quad M \delta \hat{U} \leq \rho \quad (46)$$

$$E = H_y^T Q H_y + R \quad (47)$$

$$f = H_y^T Q [\bar{Y} + G_y \delta x_k] \quad (48)$$

where $E \in \mathbb{R}^{N_p n_u \times N_p n_u}$ is a symmetric matrix known as the Hessian and $f \in \mathbb{R}^{N_p n_u}$ is a column vector usually referred as the linear term, $M \in \mathbb{R}^{2N_p n_u \times N_p n_u}$ is the constraints matrix and $\rho \in \mathbb{R}^{2N_p n_u}$ is the constraints vector, defined as:

$$M = \begin{bmatrix} I \\ -I \end{bmatrix}, \rho = \begin{bmatrix} U_{\max} - \bar{U} \\ -(U_{\min} - \bar{U}) \end{bmatrix} \quad (49)$$

In the present work, only constraints in the control input are considered. If any states' constraints are required, M and ρ must be slightly reformulated. It is worth noting that G_y and H_y , and hence E and f , are time-dependent, which is one of the main reasons why NMPC is computationally expensive.

After defining E , f , M , and ρ , the OCP can be solved using any QP solver, for example, Matlab's *quadprog* function and qpOASES [44]. Quadprog was used in this paper. Once the QP problem is solved, the new control input sequence is computed, remembering that $\hat{U} = \bar{U} + \delta U$. Only the first input from the new control input sequences is applied to the system, and the procedure is repeated at the next time step, which is known as the receding horizon scheme [65].

4. Results

In this section, we employ the proposed algorithm for predicting wave excitation moment, benchmarking its performance against several state-of-the-art methods. Specifically, we compare our algorithm with long-short term memory and auto-regressive models. To ensure a comprehensive evaluation of the state-of-the-art algorithms, testing is conducted under both regular and irregular wave conditions, encompassing six distinct group cases for each wave condition (refer to Table 3). The numerical results of the proposed moment prediction strategy applied to the benchmark Wavestar scale model simulated on WEC-Sim. The specific model parameters are reported in Table 2 (see Fig. 5).

Table 2
Model parameters for the scale model of the Wavestar device.

Hydrodynamic parameters		
Inertia of arm and float	J	1.04 kg m ²
Added inertia	J_{∞}	0.4805 kg m ²
Hydrostatic stiffness coefficient	K_{hs}	92.33 N m rad ⁻¹
Rotational linear damping	b_v	1.80 N m rad ⁻¹ s ⁻¹
Radiation moment impulse response realization		
$A_r = \begin{bmatrix} -13.59 & -13.35 \\ 8.00 & 0.00 \end{bmatrix}$	$B_r = \begin{bmatrix} 8.0 \\ 0.0 \end{bmatrix}$	
$C_r = \begin{bmatrix} 4.739 & 0.5 \end{bmatrix}$	$D_r = -0.1586$	

Table 3
Parameters for wave generation using JONSWAP spectrum. Significant wave height H_m , peak period T_p and peak enhancement factor γ .

Categories	H_m [m]	T_p [s]	γ [-]	Duration [s]
Case #1	0.0208	0.988	1.0	98.8
Case #2	0.0625	1.412	1.0	141.2
Case #3	0.1042	1.836	1.0	183.6
Case #4	0.0208	0.988	3.3	98.8
Case #5	0.0625	1.412	3.3	141.2
Case #6	0.1042	1.836	3.3	183.6

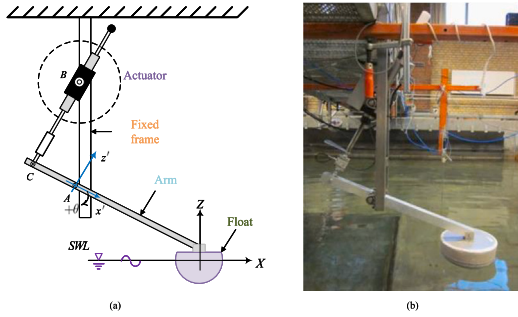


Fig. 5. The 1:20 scale laboratory representing the Wavestar prototype WEC. (a) sketch of the device with hydrodynamic and structural loads; (b) physical model in the laboratory.

4.1. Irregular waves condition

In this section, the proposed algorithms undergo testing under irregular wave conditions. The wave data is collected from the wave basin installed at Aalborg University. It is a device with a 15.7 m length, 8.5 m width, and 0.7 m depth experimental Wavestar prototype. The laboratory prototype is used as the control competition and algorithm verification. It consists of a nearly hemispherical float, mechanically hinged to a fixed reference point (commonly labeled A in Fig. 1). The power-take-off device is a linear function with the wave excitation moment, an accelerator and position sensor provide measurements of the float motion, so the position, velocity, and acceleration variables are available during the experimental process. Meanwhile, a load cell measures the PTO force applied by the linear PTO. In the Matlab Simulink model, the linear measurements are first transformed to the angular reference framework corresponding to the single degree of freedom of the WEC, which is corresponding to the hinged setpoint A. Then a linear Kalman filter is used as the wave excitation moment estimation from the measurable force, position, acceleration, and velocity.

In the later NMPC-based architecture, quality prediction is the key factor that affects the control performance. The wave excitation moment prediction is also the main task in this present analysis. Considering the available dataset as a time series of wave moment estimations, it represents the output derived from the wave force estimation block. These estimations are logged during closed-loop experiments where nonlinear real-time model predictive control is applied to maximize

WEC energy absorption. The experiments encompass six distinct random wave conditions. Each sea state is representative of the JAWSWAP spectrum, which represents the classic different case in the basin. Specifically, Case 1 and Case 4 have the same wave parameters, Case 2 and Case 5, and Case 3 and Case 6 are the same setups. Case 3 and Case 6 are the most interesting sea state since they correspond to the most frequent sea states and contain much more energy than the other 2 groups. All of the available cases are composed of irregular waves. The waves are unconditional, two-dimensional, long-crested waves, and are generated by a wave gauge whose paddles are driven by a total of 15 hydraulic pistons moving in the receding horizontal direction. All the wave excitation moment data are sampled with a 20 Hz frequency, in order to reduce the computation burden of the prediction algorithms. Due to the initial MPC being a computational burden, its calibration had to be changed. This experimental condition is also included in [11, 15,42,43], we are herein cited for the self-contained. Regarding the prediction horizon, the wave excitation force prediction suggests that the performance is superior when if the prediction can be extremely accurate up to two peak wave periods in the future, which is set as $N_p = 2 * T_p/dt$. All the simulation time is set as 200 s.

4.2. Wave excitation moment prediction results and performance analysis

The competitive algorithms such as autoregressive (AR), long-short time memory, and diffused methods are compared in the simulation experiments. Fig. 6 and Fig. 7 describe the prediction performance different under the sea case 1, where the red dot line is the optimal real value of the wave excitation moment, which defines the optimal bound of the prediction algorithm can achieve. The AR algorithm is marked as blue, and the LSTM is given in a pink dot line. The black dot line is the proposed probabilistic diffused LSTM method, and the gray area is the 95% prediction confidence interval. It can be shown that the proposed diffused achieve the optimal prediction performance, while the autoregressive prediction accuracy is lower than the proposed method. The LSTM has a good learning ability that can be used to predict the temporal information, however, the gradient cannot be learned, so the whole performance is lower than the other three methods. The convergence performance are also tested on the different optimizer, such as SGD, ASGD, Adam, AdamW, NAdam and RMSprop. The results conducted with Diffused-LSTM model are given in Fig. 8 and Fig. 9, respectively. The learning rate for all these optimizer are set as 0.001, and the momentum hyper-parameter for RMSprop and SGD are set as 0.9. The epoch is set as 500. The results shows that compared to other six optimizer, RMSprop achieves the fast convergence speed. NAdam's convergence is also satisfactory while the stability is not satisfactory. This is the acceleration result with oscillation in some special point. Compared to Adam, SGD converges slower and even the improved ASGD, the convergence speed is limited.

Table 4 summarizes the mean square error and root mean square error with different prediction approaches under the six wave conditions, it shows that the probabilistic diffused method achieves relatively low rmse, and mse, compared to the autoregressive and long-short time memory prediction. Especially in Case 3 and Case 6 waves, the mse of both two cases is 0.5833, which is lower than the 0.7257 under Case 3, and 0.7009 under Case 6, respectively. The rmse of the proposed diffused LSTM also achieve 0.7650 rmse under the sea state 6, which improved the performance of the wave excitation moment by a satisfactory margin. Compared to TCN [67], eESN [68] shows a better prediction performance in the 6 different waves, while Transformer outperforms other prediction methods, which is reasonable under the big volume training wave datasets. NBbeats [69] and DeepAR [70] also show a good prediction ability in wave excitation moment prediction, achieve 0.6685 in Case 6. The results demonstrate that under the 6 different wave cases, the LSTM has better performance, which is caused by the time series information are effectively extracted.

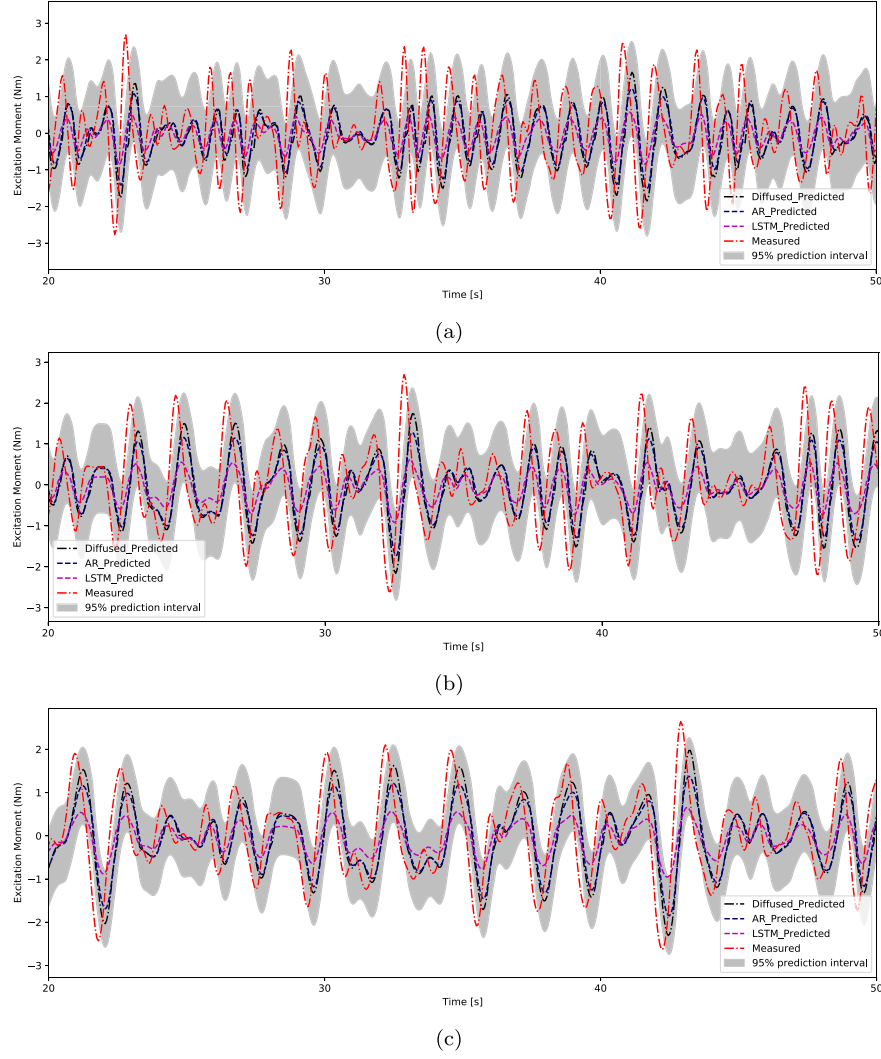


Fig. 6. Six wave case comparisons with the different wave excitation moment prediction algorithms. (a)–(f) are the normalized wave excitation moment prediction under sea cases 1–3.

Table 4

The rmse index of the different wave excitation prediction under the different wave cases.

Methods	Index	Case #1	Case #2	Case #3	Case #4	Case #5	Case #6
AR	<i>MSE</i>	1.4379	0.9913	0.7119	1.3489	0.9152	0.6864
	<i>RMSE</i>	1.1991	0.9956	0.8437	1.1614	0.9566	0.8285
LSTM	<i>MSE</i>	1.3919	1.0265	0.7257	1.4015	0.9485	0.7009
	<i>RMSE</i>	1.1798	1.0132	0.8519	1.1838	0.9739	0.8371
TCN [71]	<i>MSE</i>	1.2876	1.0378	0.6238	1.3278	0.8863	0.6534
	<i>RMSE</i>	1.1347	1.0187	0.7898	1.1523	0.9414	0.8083
edESN [68]	<i>MSE</i>	1.2156	1.0546	0.7488	1.4329	0.9003	0.9210
	<i>RMSE</i>	1.1025	1.0269	0.8653	1.1970	0.9488	0.9596
Transformer	<i>MSE</i>	0.8007	0.9204	0.8751	0.9193	0.8856	0.6047
	<i>RMSE</i>	0.8948	0.9593	0.9354	0.9588	0.9410	0.7776
NBbeats [69]	<i>MSE</i>	1.4027	1.1128	0.9872	1.5063	1.3118	0.8926
	<i>RMSE</i>	1.1843	1.0548	0.9935	1.2253	1.1453	0.9447
DeepAR [70]	<i>MSE</i>	0.9006	0.7290	0.7291	0.9363	0.7433	0.6685
	<i>RMSE</i>	0.9489	0.8538	0.8538	0.9676	0.8621	0.8176

Table 5 summarizes the result of different diffusion models. It shows that the diffusion-based model outperforms the traditional prediction methods. We also compared the CNN-diffusion prediction, MLP-Diffusion model, and the proposed LSTM prediction model. LSTM-Diffusion model achieves the superior prediction accuracy and MSE and RMSE are 0.5853, and 0.7650, respectively.

4.3. Control result and analysis

With the idea of having a reference point for each state, simulations are carried out using NMPC with different wave excitation moment prediction strategies. Subsequently, the assessment of prediction Root Mean Square Error (RMSE), mean power, and absorbed energy will

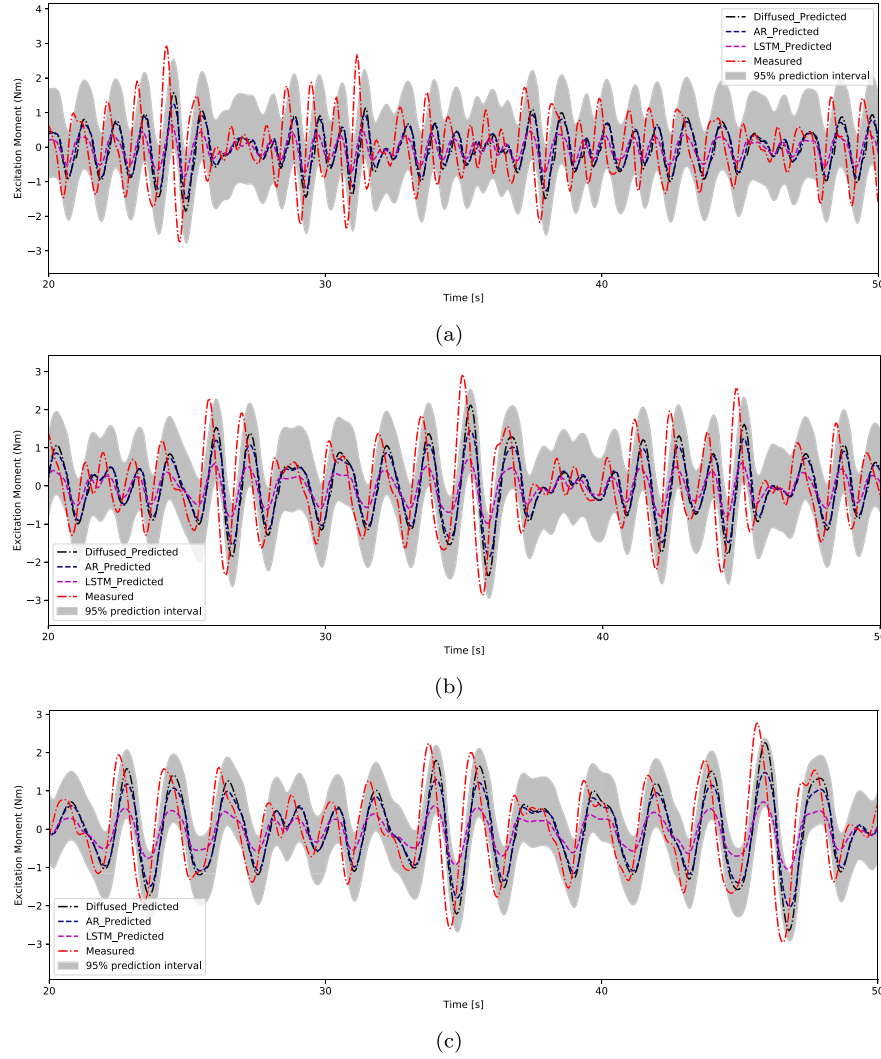


Fig. 7. Six wave case comparisons with the different wave excitation moment prediction algorithms. (a)–(f) are the normalized wave excitation moment prediction under sea cases 4–6.

Table 5

The ablation study for diffusion-based predictions with the different wave cases.

Methods	Index	Case #1	Case #2	Case #3	Case #4	Case #5	Case #6
CNN-Diffusion	<i>MSE</i>	1.0812	1.0628	0.9701	1.2389	1.0348	0.8142
	<i>RMSE</i>	1.0398	1.0309	0.9849	1.1130	1.0172	0.9499
MLP-Diffusion	<i>MSE</i>	0.9921	0.9172	0.9092	0.8217	0.8668	0.7763
	<i>RMSE</i>	0.9960	0.9577	0.9535	0.9064	0.9310	0.8810
LSTM-Diffusion	<i>MSE</i>	0.8997	0.7019	0.5833	0.8544	0.6453	0.5853
	<i>RMSE</i>	0.9485	0.8378	0.7637	0.9243	0.8033	0.7650

afford a comprehensive and intuitive elucidation of the performance of the proposed Diffused-LSTM model. Figs. 13–15 gives the numerical simulation results on the different sea states, Case 4, Case 5, and Case 6 benchmarks, respectively. Specifically, Fig. 13(a) presents the wave elevation, which is the main source of energy, and (b) is the corresponding wave excitation moment, which has been scaled by the normal transformation. In Fig. 13(b), the wave excitation moment prediction of the AR-based algorithm, the proposed diffused long-short time memory prediction, and the real measured wave excitation moment value. Under the 95% prediction interval, the diffused wave excitation moment gives a better fitting about the wave excitation moment prediction, which means the higher power potential in the later PTO unit. Figs. 13(c)–(d) are the control input for the PTO devices and their simultaneous power, it can be analyzed that in the subprime interval of [43–60], the diffused long-short time memory method is

more accurate than the AR-based predictor. Notability, the simultaneous energy absorption is corresponding proportional incrementing with the wave elevation, also the wave excitation moment. Fig. 14 studied the simulation under Case 5, compared to Case 4, it has higher wave peaks and periods, higher wave elevation, and wave excitation moment, thus more energy can be extracted from the sea wave conditions. Some similar results can be drawn from Case 4, the diffused-based long-short time memory implements a long temporal moment prediction thus a higher energy absorption efficiency. Fig. 15 describes the results under the sea state 6, which also has the highest wave elevation and wave peak. Fig. 15(b) and (d) depict that the prediction in the sub-time domain interval, the diffused long-short time memory achieves higher and more accurate prediction, the output of the simultaneous power is higher, yet a more oscillation is given, which is also corresponding with our intuition.

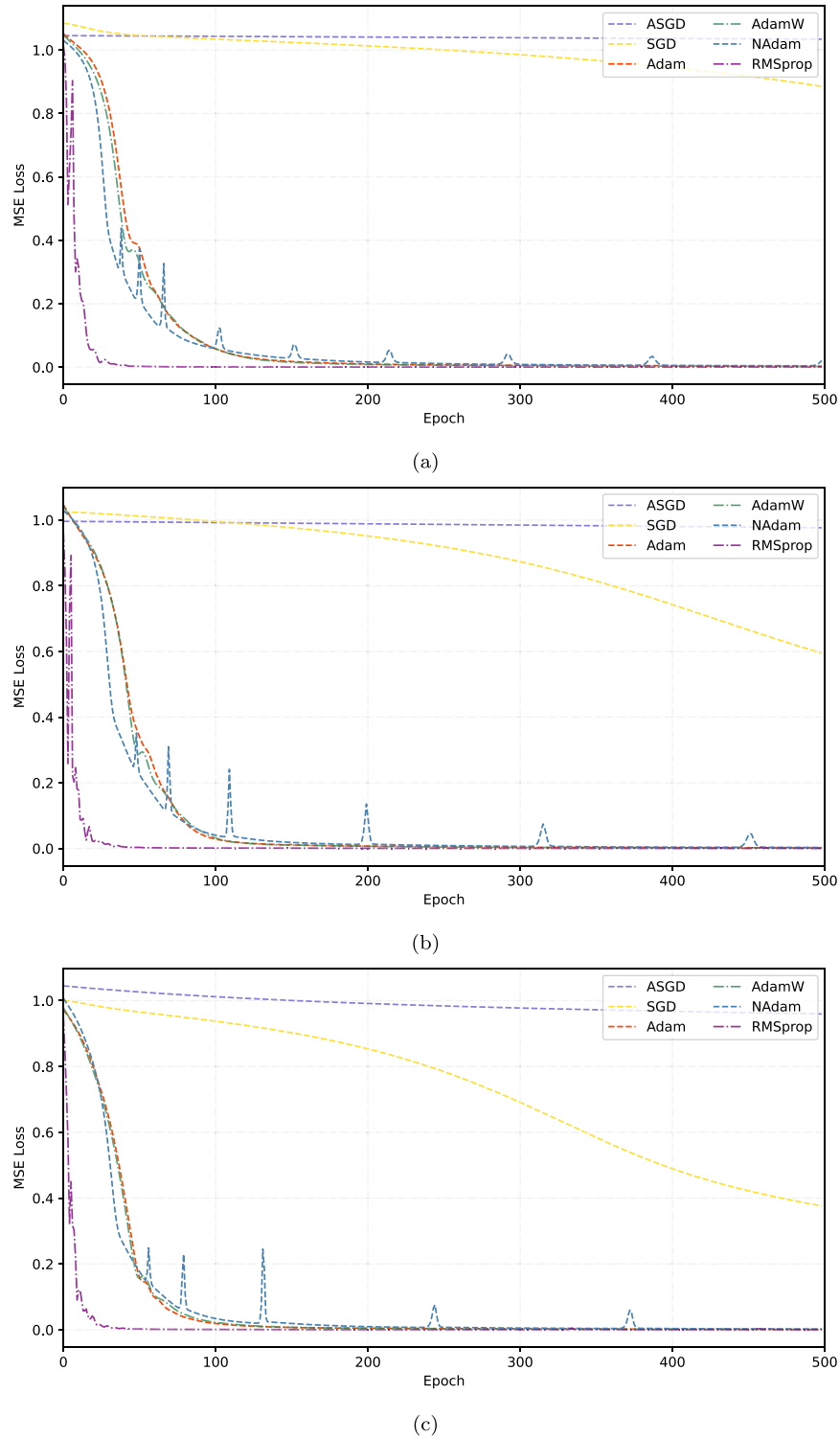


Fig. 8. Six wave case comparisons on Diffused-LSTM with the different optimizer. (a)–(c) belong to Cases 1–3, respectively.

For a comprehensive ablation analysis, Table 6 consolidates the numerical results across six distinct sea states, encompassing 4-fold cross experiments conducted under random seeds 2.3, 3.3, 4.3, and 5.3. To ensure a fair comparison, all controllers for the six wave conditions are rooted in the same Nonlinear Model Predictive Control (NMPC), with random seeds set to 3.3, 2.3, 4.3, and 5.3, respectively. Some information can be gained from Table 6, firstly, the mean power and energy absorptions are the direct reflection of the wave excitation moment prediction performance and controller effectiveness. Under Case 1, the

proposed diffused-based long-short time memory achieves 5.16 J, and a mean power of 24.64 mW, compared to AR-based prediction, 3.95 and 18.93 respectively. Similarly, case 6 suggests that the diffused long-short time memory for the wave excitation moment extracts 183.04J energy, with a mean power of 865.99 mW, which is higher than 76.06, 355.40, AR-prediction harvested. Another observation from Table 6 is that the optimal random seeds are that the performance is better to set the random seeds as 3.3, compared to 2.3, 4.3, and 5.3. This is an experimental trivial result that helps to reduce the experimental error.

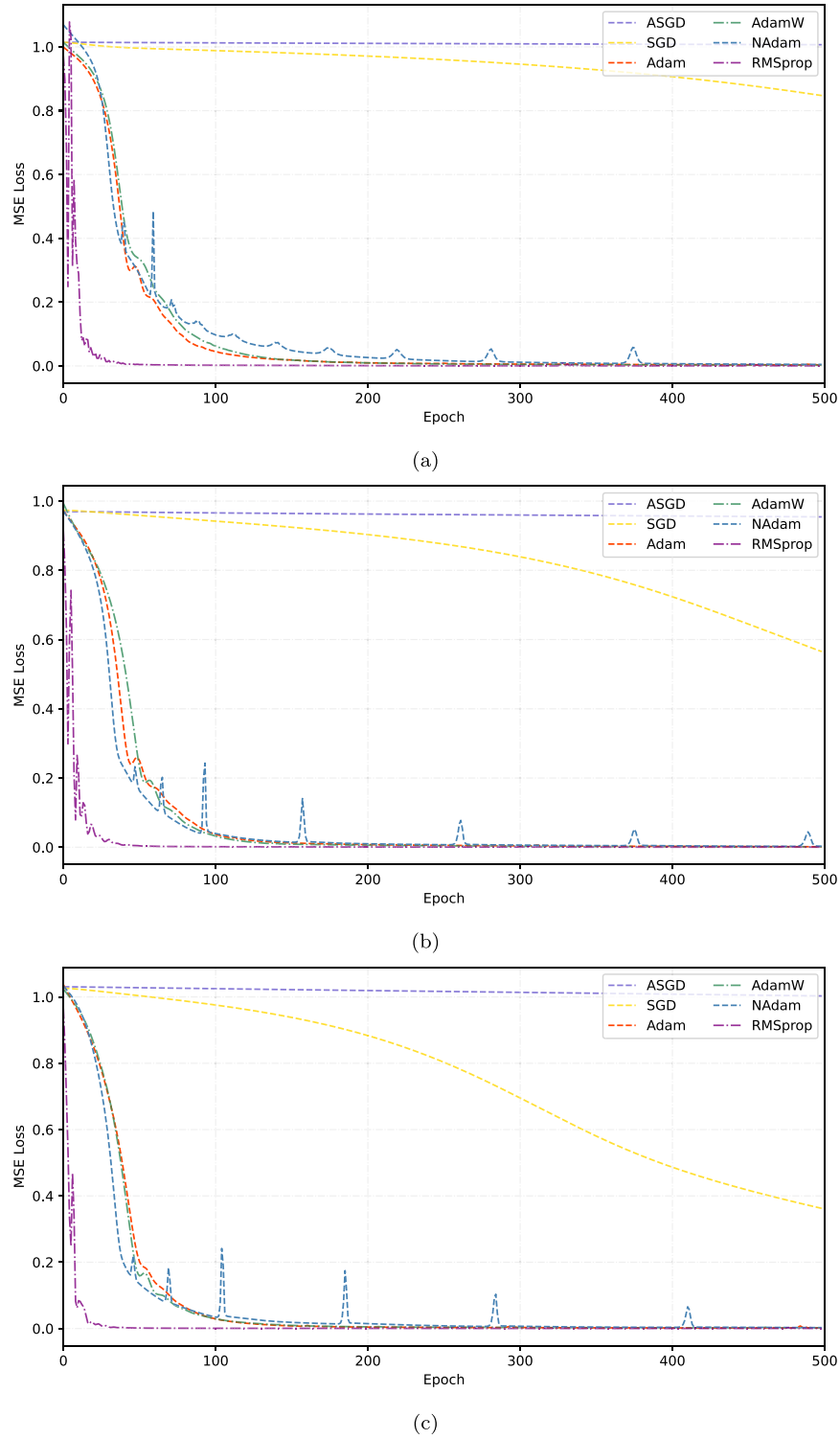


Fig. 9. Six wave case comparisons on Diffused-LSTM with the different optimizer. (a)–(c) belong to Cases 4–6, respectively.

Random seeds 3.3 achieves the best performance and with the highest energy and mean power in most cases, yet in Case 2, the power with the setting of 2.3, which achieves 323.93 mW, which is higher than 321.59, when the random seeds set as 3.3. Henceforth, in subsequent experiments, we opt for random seeds set at 3.3.

Finally, the total energy absorption with different prediction methods is given in Figs. 10–12. As shown in Fig. 10, 4 different curves are

given, the red represents the real wave excitation moment, which is also the theoretical optimal value for the prediction. In such conditions, the wave excitation moment equals the real measured value, without any residual error. Other comparison methods are the LSTM-based, AR-based, and our proposed diffused LSTM prediction method. In Fig. 10, AR-based prediction achieves 5.27 J, while the diffused LSTM extracts 5.49 J energy, with an increase of 22%, while the LSTM-based

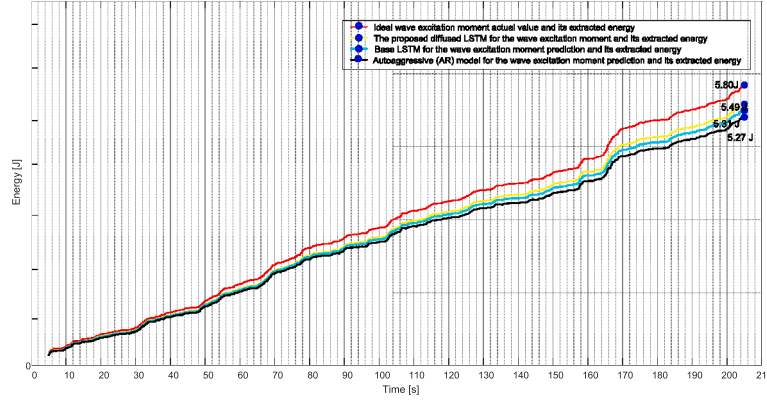


Fig. 10. The SS4 Sea state for the whole energy absorption performance comparison.

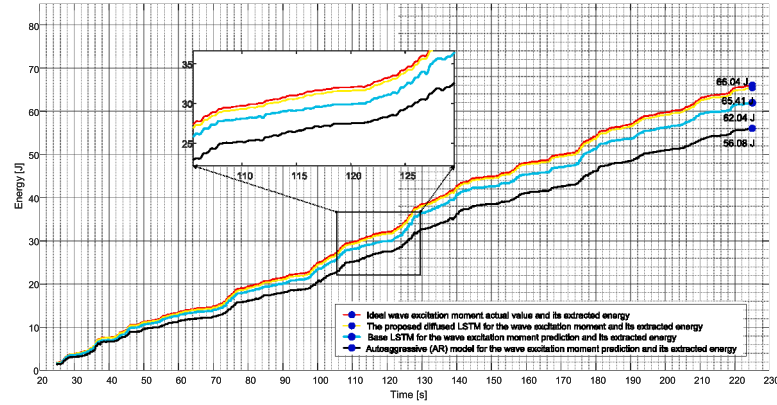


Fig. 11. The SS5 Sea state for the whole energy absorption performance comparison.

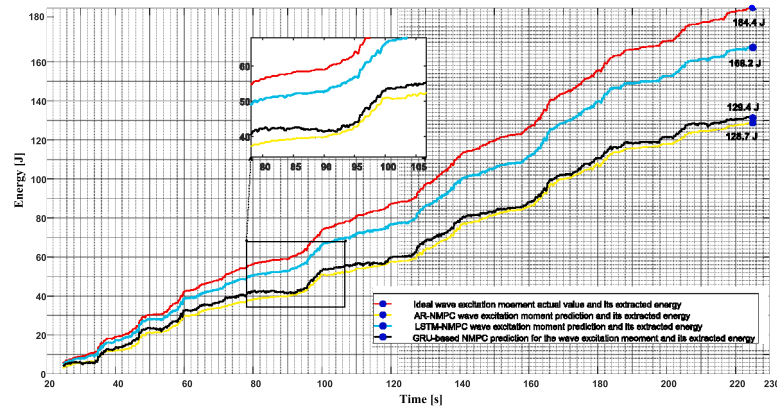


Fig. 12. The SS6 Sea state for the whole energy absorption performance comparison.

prediction extracts 5.31 J, which is higher than the AR-based. Fig. 10 shows that the proposed diffused LSTM achieves superior performance, compared to the LSTM base and autoregressive model. Fig. 11 gives the results under Case 5, the subfigure which represents the local information, the prediction tendency corresponds with the LSTM-based wave excitation moment prediction. In addition, the proposed diffused LSTM harvests 65.64 J energy, an increase of 16.64%, compared to AR's 56.08. Fig. 12 is the most intuitive description of the proposed diffused LSTM prediction performance, the total energy extraction series are

184.4 J, 168.2 J, 129.4 J, and 128.7 J respectively. Compared to AR-based and LSTM-based wave excitation moment prediction, the diffused LSTM improves by 30.6%, and 29.9%, respectively. The gap with the optimal real prediction is only 16.2 J, which is a satisfactory and feasible prediction. What is more, the analysis of Fig. 12 shows that the AR-based prediction performance is synonymous with the LSTM-based wave excitation moment prediction, while the diffused LSTM improved by a large margin.

Table 6

Prediction performance comparison with different Randseed for the control and energy absorption. All the experiments are based on the NMPC-control strategy.

AR prediction/ Diffused-LSTM prediction	Random seeds	Case #1	Case #2	Case #3	Case #4	Case #5	Case #6
Power [mW]	3.3	18.93	154.02	377.53	19.51	146.79	355.40
	2.3	18.86	149.72	374.53	21.16	143.21	366.34
	4.3	17.38	141.96	377.76	19.27	153.87	376.63
	5.3	17.55	131.97	349.03	17.65	148.39	370.37
Mean Power	–	18.18	144.42	369.71	19.40	148.07	367.19
Energy absorption [J]	3.3	3.95	31.66	77.92	4.05	30.21	73.41
	2.3	3.86	31.12	77.19	4.39	30.04	75.17
	4.3	3.67	29.31	77.88	4.00	31.36	78.36
	5.3	3.70	27.94	71.78	3.69	30.23	77.28
Mean Energy absorption	–	3.80	30.01	76.19	4.03	30.46	76.06
Power	3.3	24.64	321.59	963.77	26.06	320.84	891.86
	2.3	23.20	323.93	966.43	21.26	318.75	877.10
	4.3	23.77	316.37	960.62	24.56	316.96	881.29
	5.3	23.60	320.68	955.23	24.24	310.77	813.72
Mean Power	–	23.80	320.64	961.51	24.03	316.83	865.99
Energy absorption	3.3	5.16	66.13	199.13	5.44	66.04	184.38
	2.3	5.12	66.36	199.13	5.39	65.59	182.93
	4.3	5.09	65.19	197.88	5.04	66.33	183.71
	5.3	4.98	64.94	198.78	5.22	64.38	181.14
Mean Energy absorption	–	5.09	66.04	198.73	5.27	65.59	183.04

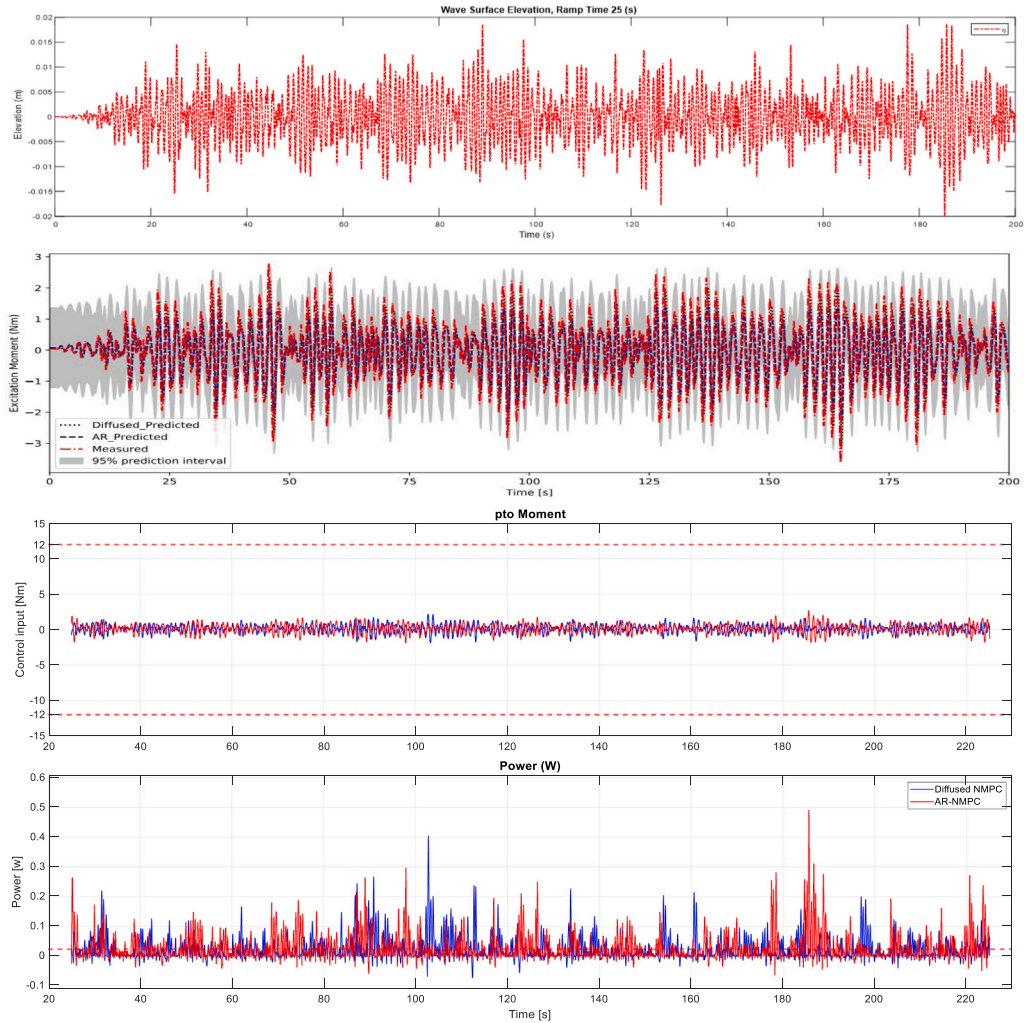


Fig. 13. The SS4 Sea state for the whole performance comparison; (a) wave elevation; (b) wave excitation moment prediction; (c) Control input for PTO; (d) Simultaneous power.

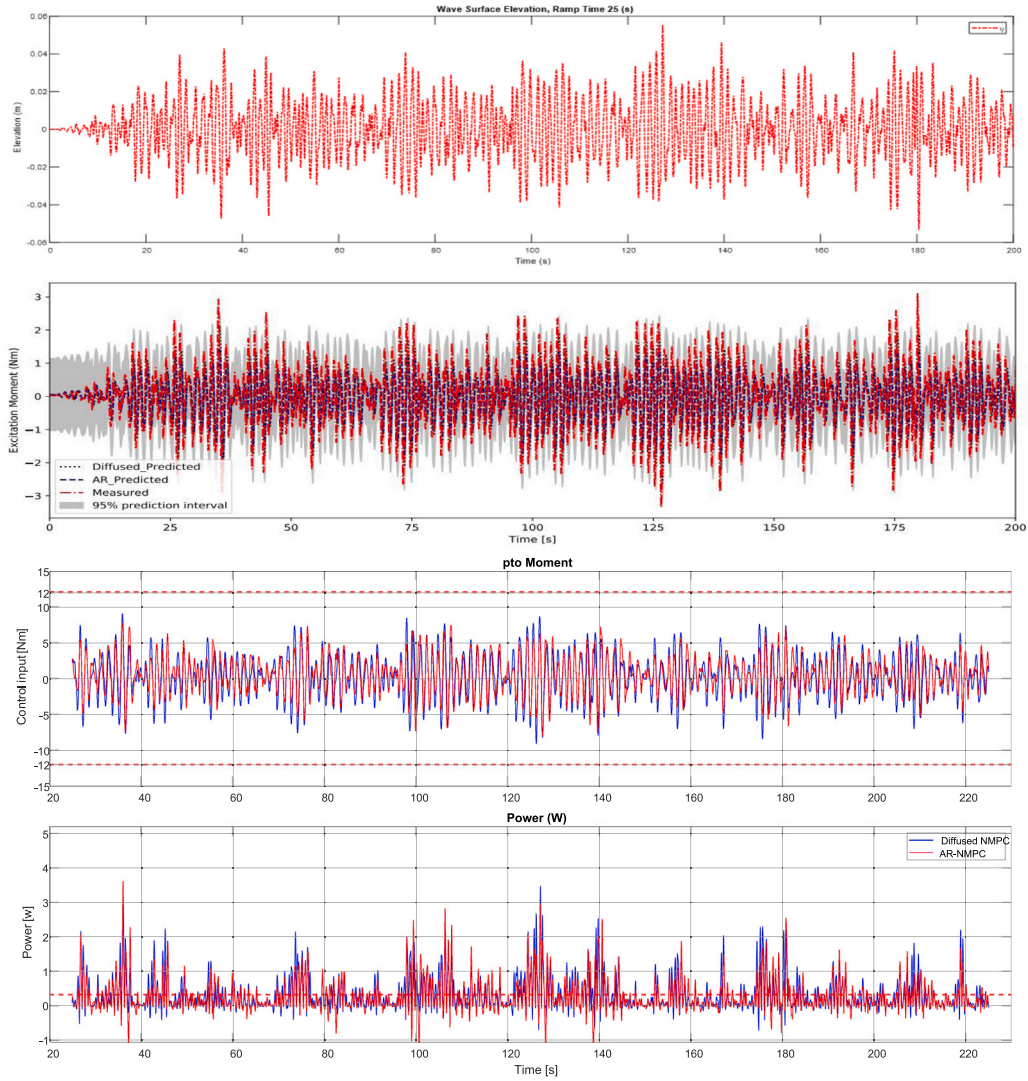


Fig. 14. The SS5 Sea state for the whole performance comparison; (a) wave elevation; (b) wave excitation moment prediction; (c) control input for PTO; (d) Simultaneous power.

5. Conclusion

This paper introduces a pioneering wave excitation moment prediction algorithm rooted in probabilistic diffused long-short time memory. Leveraging long-short time memory as its fundamental building block, the proposed method integrates a diffused probabilistic network, facilitating the extraction of hidden features with temporal instant scale. Additionally, the efficacy of the prediction is rigorously evaluated through the application of the nonlinear model predictive control method. Comparative analysis against AR-based and neural-based prediction methods reveals the superior performance and reduced Mean Squared Error (MSE) of the proposed diffused probabilistic approach under six-wave condition cases. Notably, in Case 4, Case 5, and Case 6, the energy absorption and mean simultaneous values exhibit marked improvements of 22%, 16%, and 30%, respectively. Furthermore, a noteworthy contribution of this work lies in the integration of the presented probabilistic prediction into a comprehensive control framework. This integration establishes a unified full-scale prediction-and-control framework, enhancing the practical applicability and versatility of the proposed methodology.

6. Discussion

One of advantages of proposed framework is the wave prediction accuracy, the improvement in accuracy helps the control performance. The second advantage of the presented method is the uncertainty consideration. A unified prediction-and-control full-scale framework is proposed, which is the last contribution. The proposed diffused LSTM-based prediction framework can be extended to other domains and improve the generalization of the prediction. Especially some temporal time-scale systems with advanced control performance. The proposed method also achieves some similar prediction results to the generative adversarial network and Bayesian neural network, yet yields a more complex hidden representation. One limitation of the presented framework is the sampling from the Markovian sampling, which strictly relies on the original data manifold distribution. Another limitation is that the presented model is highly time-sequence model, the historical observation plays a vital importance for the model performance. A possible direction to explore our proposed method further is to generalize the diffusion process with faster sampling, which can be implemented by a class of non-Markovian processes. On the other hand, the wave excitation moment is the basis for the wave energy converter, some other advanced control strategies should be combined for maximizing the energy harvest in the wave-energy devices.

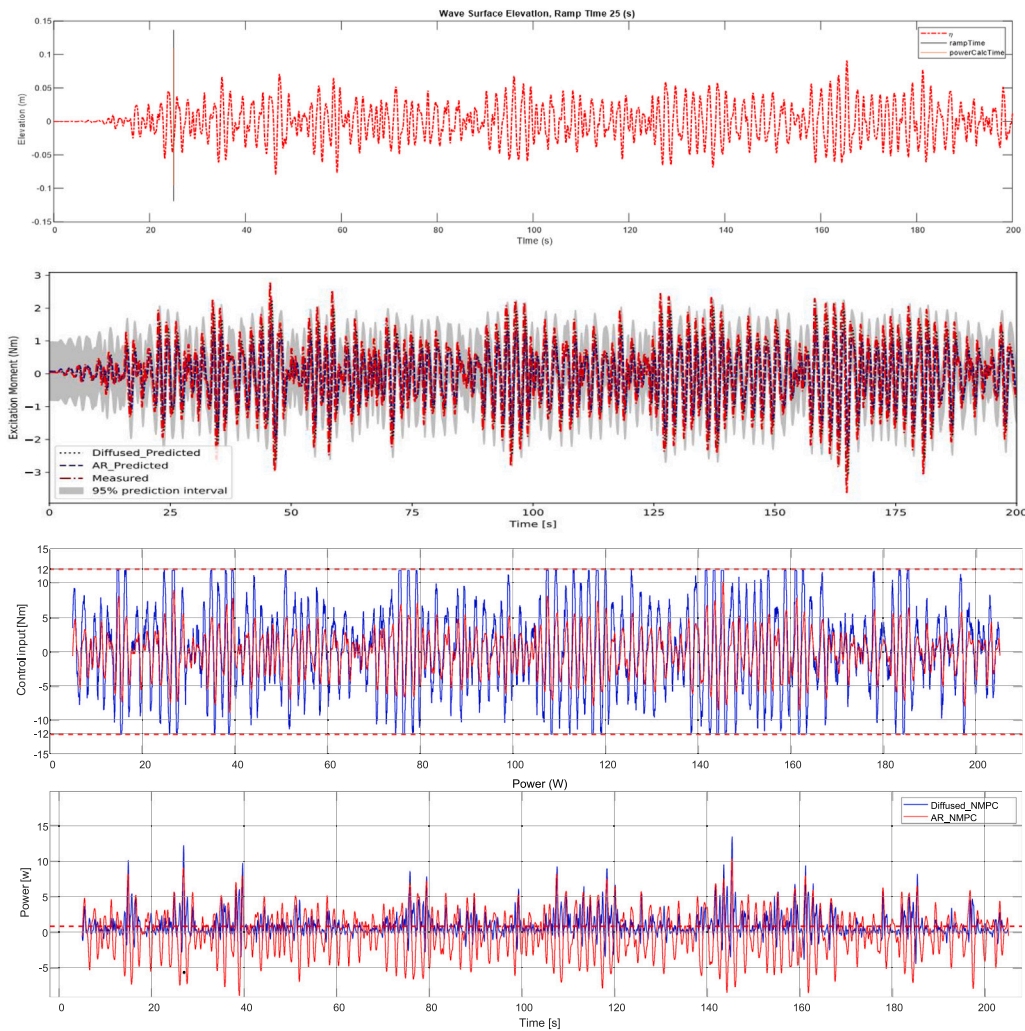


Fig. 15. The SS6 Sea state for the whole performance comparison; (a) wave surface elevation; (b) wave excitation moment prediction; (c) control input for PTO; (d) simultaneous power. The similar simulation prediction results obtained in case 1, 2, and 3 wave conditions, so not presented here.

For future consideration, we will focus on the study of robust control with the model uncertainty, specifically, the local exploration and exploitation for the real-time switching of the different control methods which could further improve the energy efficiency and the performance of the prediction and control.

CRediT authorship contribution statement

Yongxiang Lei: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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